

Nahyuk Lee

POSITION	Research Scientist VUNO Inc.
CONTACT INFORMATION	VUNO Inc. 8F, Simnonhyeon Tower 479, Gangnam-daero, Seocho-gu, Seoul, 06541 Republic of Korea. E-mail: nahyuk.lee@postech.ac.kr nahyuk.lee@vuno.co Homepage : https://nahyuklee.github.io/
CITIZENSHIP	The Republic of Korea
BIRTHDAY	Jan 13th, 2000.
RESEARCH INTERESTS	Computer vision and machine learning , in particular, the problems of relational computer vision: visual correspondence, 3D shape matching/assembly, and equivariant learning. Medical AI and clinical research , in particular, the problems of 3D medical image (CT/MRI) computing: deformable registration, nodule/siderosis segmentation, and computer-assisted diagnosis.
EDUCATION	Pohang University of Science and Technology (POSTECH) , Pohang, Korea <i>M.S. in Computer Science and Engineering</i> Feb 2023 - Feb 2025 • Thesis : <i>Combinative Matching for Geometric Assembly</i> • Advisor : Prof. Minsu Cho • Admission with the highest distinction (Bang Sung-Yang Fellowship) Chung-Ang University , Seoul, Korea <i>B.S. in Computer Science and Engineering</i> Mar 2019 - Feb 2023 • Advisor : Prof. Junseok Kwon • Cumulative GPA : 4.31/4.5 (Summa Cum Laude, 1st in Class of 2023) Hyundai Chung-Un High School , Ulsan, Korea <i>Natural Science Track</i> Mar 2015 - Feb 2018
WORK EXPERIENCE	VUNO Inc. , Seoul, Korea <i>Research Scientist (Military Obligation)</i> Jan 2025 - now • Working with Dr. Sunghoon Joo (R&D Lead). • Researching 3D computer vision for lung CT and brain MRI analysis: — <i>Geometric deep learning</i>: Researching group equivariant brain segmentation for computer-assisted ARIA-H diagnosis. Collaborating with Asan Medical Center (Jun 2025 - now). — <i>Visual correspondence</i>: Researched point-based efficient deformable lung CT registration [Appl. Sci. 2025] (Jan 2025 - May 2025). • Managed volumetric super-resolution and solid portion segmentation modules in VUNO Med[®]-LungCT AI[™] (Jan 2025 - Jun 2025).
RESEARCH EXPERIENCE	Computer Vision Lab, POSTECH , Pohang, Korea <i>Graduate Research Assistant</i> Jul 2022 - Dec 2024 • Advised by Prof. Minsu Cho • Researched learning visual correspondence and its applications, with a particular focus on 3D point cloud registration, geometric shape assembly, and equivariant learning: — <i>Group-equivariant learning</i>: Researched SE(3)-equivariant <i>combinative</i> matching for 3D geometric shape assembly [ICCV 2025 Highlight]. — <i>Visual correspondence</i>: Researched computational efficient 3D point cloud matching for point cloud registration and geometric shape assembly [ICML 2024; IPIU 2024 Oral].

- Collaborated with [VGI Lab, SNU](#) (Kanghee Lee, [Prof. Jaesik Park](#)).

Undergraduate Research Fellowship

Jan 2022 - Feb 2022

- Advisor : [Prof. Jaesik Park](#)
- Researched point cloud augmentation and sampling for robust 3D shape classification.
- Collaborated with a student at CUHK (Hansheng Guo).

Computer Vision Lab, Korea University, Seoul, Korea

Research Intern (full-time)

Mar 2021 - Nov 2021

- Advisor : [Prof. Sangpil Kim](#)
- Researched audio-guided image manipulation [NeurIPSW 2021], and event-based photometric stereo.

Computer Graphics Lab, Chung-Ang University, Seoul, Korea

Research Intern (full-time)

Dec 2019 - Feb 2021

- Advisor : [Prof. Sanghyun Seo](#) and Prof. Kyunghyun Yoon
- Researched automated 3D reconstruction application [JDCS 2020], and affective computing [CSCI 2020 ; JDCS 2021].

PUBLICATIONS

* indicates equal contribution.

[C4] **Nahyuk Lee***, Juhong Min*, Junhong Lee, Chunghyun Park, Minsu Cho, “Combinative Matching for Geometric Shape Assembly,” in *Proceeding of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, **Highlight (263/11,152 $\approx$ 2.35%)**

- Joint presentation at [Structural Priors for Vision \(SP4V\) Workshop](#), ICCV 2025
- Invited oral session at [French-Korean Workshop on AI, Network & Communication 2024, Korea AI Summit 2024](#), and POSTECH AI DAY 2024
- Extension of my Master’s Thesis (Committees : Minsu Cho, Jungseul Ok, and Seungyong Lee)

[C3] Doohyun Park, **Nahyuk Lee**, Sungjoo Lim, “Attention-based Interpretable Deep Learning with Radiomic Features for Pulmonary Nodule Classification,” in *Medical Imaging and Deep Learning (MIDL)*, Short Paper, 2025

[C2] **Nahyuk Lee***, Juhong Min*, Junha Lee, Seungwook Kim, Kanghee Lee, Jaesik Park, Minsu Cho, “3D Geometric Shape Assembly via Efficient Point Cloud Matching,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2024

- **Best Paper** at [IPIU 2024](#) (1st of 311 papers $\approx$ 0.3%) with **oral** presentation
- Invited poster session at [Samsung AI Forum \(SAIF\) 2024](#)

[C1] Taemin Lee, **Nahyuk Lee**, Sanghyun Seo, Dongwann Kang, “A Study on the Prediction of Emotion from Image by Time-flow depend on Color Analysis,” in *Proceeding of the International Conference on Computational Science and Computational Intelligence (CSCI)*, 2020

[W1] Seung Hyun Lee, **Nahyuk Lee**, Chanyoung Kim, Wonjeong Ryoo, Jinkyu Kim, Sang Ho Yoon, Sangpil Kim, “Audio-Guided Image Manipulation for Artistic Paintings,” in *NeurIPS Workshop on Machine Learning for Creativity and Design (NeurIPS Workshop)*, 2021

[J3] **Nahyuk Lee**, Taemin Lee, “Bi-directional Point Flow Estimation with Multi-scale Attention for Deformable Lung CT Registration,” in *Applied Sciences*, 2025

[J2] **Nahyuk Lee**, Taemin Lee, “The Study for Emotion Analysis from Images depend on Machine Learning Models,” in *Journal of Digital Contents Society (JDCS)*, 2021

[J1] **Nahyuk Lee**, Kyungtaek Lee, Youngsup Park, Sanghyun Seo, Taemin Lee, “Development of 3D Reconstruction and Object Recognition Model using Video,” in *Journal of Digital Contents Society (JDCS)*, 2020

ACADEMIC
SERVICES

Reviewer of international conferences

International Conference on 3D Vision (**3DV**)
International Conference on Learning Representations (**ICLR**)
LSRW Workshop, Conference on Robot Learning (**CoRLW**)
WiCV Workshop, Asian Conference on Computer Vision (**ACCVW**)

Reviewer of international journals

Transaction on Pattern Analysis and Machine Intelligence (**TPAMI**)

IF : 20.8

HONORS AND
AWARDS

Travel Grant, IEEE/CVF International Conference on Computer Vision (ICCV), 2025
- Waived registration fees with partial scholarship from ICCV 2025, **\$1,100** (total)

[18th Kwanjeong Educational Foundation Fellowship](#), KEF, 2023
- Winner, received **\$17,000** during 2 years of my graduate studies

[Bang Sung-Yang Graduate Fellowship](#), POSTECH, 2023
- Admission with highest distinction, **\$1,200**

[16th Kwanjeong Educational Foundation Fellowship](#), KEF, 2021
- Winner, received **\$16,500** during 2 years of my undergraduate studies

University Innovation Support Project Scholarship, UISPC, 2021
- Research fellowship, **\$750**

SW Specialization Scholarship, CAU, 2020 - 2022
- Research fellowship, **\$4,200** (total)

Da Vinci Scholarship, CAU, 2019
- Admission with highest distinction, **\$7,250**

Dean's List, CAU, 2019 - 2021
- 2019 Fall, 2020 Spring, 2020 Fall, 2021 Fall (4 times)

Awards

- [Best Paper Award](#) (**Grand Prize**, 1st of 311 papers $\approx$ **0.3%**),
36th Workshop on Image Processing and Image Understanding (IPIU), 2024
- Excellence Prize in Research Report,
Innovation Center for Engineering Education, CAU, 2022
- Excellence Prize in Project,
Davinci Software Institute, CAU, 2021
- Best Paper Award,
Digital Contents Society, 2020
- Excellence Award in Project,
Innovation Center for Engineering Education, CAU, 2020
- Best Paper Award (top 10% papers),
International Next-generation Convergence Technology Association (INCA), 2020
- Best Paper Award (2nd Prize),
Humanities Research Institute, CAU, 2020

TEACHING
EXPERIENCE

Pohang University of Science and Technology (POSTECH)

Teaching Assistant

Feb 2024 - Mar 2024

- CS537/AIGS537 - Artificial Intelligence & Data Science (2024 Spring)

Ministry of National Defense, Republic of Korea

Instructor

Nov 2022 - Dec 2022

- Conducted lectures on AI projects for South Korean soldiers and short-term service officers
- Collaborated with *Elice, Inc.*

Samsung Electronics Co., Ltd*Instructor***Aug 2022 - Oct 2022**

- Conducted lectures on data science and machine learning for newcomers (DS division)
- Collaborated with *Elice, Inc.*

Chung-Ang University*Teaching Assistant***Sep 2020 - Dec 2022**

- GE 56423 - Computational Thinking and AI Literacy (2022 Fall)
- GE 52431 - Understanding SW Programming for Problem Solving (2020 Fall, 2021 Spring/Fall, 2022 Spring)
- GE 51368 - Computational Thinking and Problem Solving (2021 Spring/Fall, 2021 Spring/Fall)
- CSE 17086 - Discrete Mathematics (2021 Fall)

Korea Scholar's Conference for Youth (KSCY)*Facilitator & Tutor***Jun 2019 - Feb 2021**

- Facilitator : 12th-13th KSCY computer engineering track camp facilitator
- Tutor : 15th KSCY computer engineering track tutor

LEADERSHIP

Graduate School of Dept. of CSE, POSTECH*President of Student Council***Feb 2024 - Feb 2025**

- President of graduate school council of Department of Computer Science & Engineering, POSTECH.

Kwanjeong Educational Foundation Fellowship*Vice President***Jul 2023 - Jul 2024**

- Vice president of 18th Kwanjeong Educational Foundation Fellowship mensches.

Chung-Ang University Artificial Intelligence Society (CUAI)*3rd - 4th President***Mar 2020 - Feb 2022**

- 3rd (Mar 2020 - Feb 2021) and 4th (Mar 2021 - Feb 2022) president of the student AI society at Chung-Ang University, CUI.

INVITED
PRESENTATIONS

LLMs as Research Assistants: A Practical Guide from My Latest Papers, VUNO R&D Seminar, Seoul, Korea, Jun. 2025

3D Geometric Shape Assembly via Efficient Point Cloud Matching, Samsung AI Forum (**SAIF**) 2024, Suwon, Korea, Nov. 2024

Combinative Matching for Shape Assembly, Samsung Global Technology Symposium (**GTS**) 2023, Seoul, Korea, Mar. 2023

LANGUAGE
SKILLS

Korean (native), English (fluent)

REFEREES

Minsu Cho, Ph.D. (Academic Advisor)

Mu-Eun-Jae (無垠齋) Endowed Chair Professor

Associate Professor, Pohang University of Science and Technology (POSTECH)

Jaesik Park, Ph.D. (Coworker)

Assistant Professor, Seoul National University

Juhong Min, Ph.D. (Coworker)

Senior Researcher, Samsung Research America (SRA)

Taemin Lee, Ph.D. (Coworker)

Assistant Professor, Kangwon National University